Kick-off Diagnostic · L2-5 — Completing the Square

Grade 10

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (slides 2 and 3 of the pre-session must say this, so items 2 and 6 test it).
The one idea: To complete the square on $x^2 + bx$, add $\left(\frac{b}{2}\right)^2$: to **both sides** of an equation, or add **and** subtract it in an expression.
The mistake most students make: Adding 9 to the left side only in $x^2 + 6x = 7$, or adding b^2 instead of $\left(\frac{b}{2}\right)^2$.
Readiness items 3, 4, 5, 7, 8 test: expanding $(x \pm k)^2$, halving negative numbers, reading vertex form (Lesson 2-1).
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All six stations, including the checks
B. I watched and read but skipped the checks
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

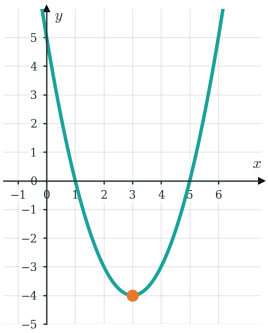
In the pre-session, what is added to $x^2 + bx$ to complete the square?

✓ **A.** $\left(\frac{b}{2}\right)^2$
B. b^2
C. $\frac{b}{2}$
D. $2b$

Signal. Wrong → did not watch station 4. Picking b^2 (B) = skipped the halving.

3 FILL IN THE BLANK · 30 s — READINESS

The graph of $y = (x - 3)^2 - 4$ is shown. Its minimum value is $y =$ _____



Accept: -4

Signal. 3 → read h instead of k; 4 → sign of k dropped. Vertex form (Lesson 2-1) needs a quick re-teach.

4 CHECKBOX · 30 s — READINESS

Select the **two** perfect square trinomials.

✓ **A.** $x^2 + 10x + 25$
B. $x^2 + 8x + 8$
✓ **C.** $x^2 - 6x + 9$
D. $x^2 - 4x + 8$

Signal. Choosing $x^2 + 8x + 8$ or $x^2 - 4x + 8$ → the last term is not $\left(\frac{b}{2}\right)^2$.

5 MATCH · 30 s — READINESS

Match each expression with the number that completes its square.

$x^2 + 8x$	→	16
$x^2 - 10x$	→	25 (shown shuffled in Wayground)
$x^2 + 2x$	→	1

Signal. 64, 100 or 4 → squared b without halving. Negative b missed → halving a negative.

6 MULTIPLE CHOICE · 30 s **PRE-SESSION CHECK**

The worked example turned $x^2 + 6x = 7$ into $(x + 3)^2 = 16$. Which wrong move did it warn about?

✓ **A. Adding 9 to the left side only**
B. Subtracting 9 from both sides
C. Adding 6 to both sides
D. Halving the 7

Signal. The pre-session's "mistake most students make". Wrong → skimmed station 5.

7 REORDER · 30 s **READINESS**

Put the steps for solving $x^2 + 6x - 7 = 0$ in order.

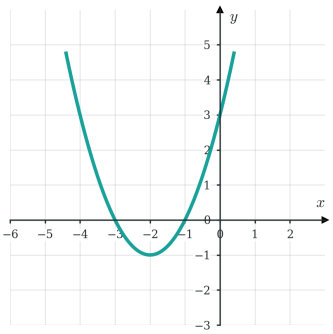
Correct order: $x^2 + 6x = 7 \rightarrow x^2 + 6x + 9 = 16 \rightarrow (x + 3)^2 = 16 \rightarrow x = 1$ or $x = -7$

Signal. Wrong → the equation routine (add to both sides) is missing.

8 DROPDOWN · 30 s **READINESS**

The graph shows $y = x^2 + 4x + 3$. Which is its vertex form?

✓ **A.** $y = (x + 2)^2 - 1$
B. $y = (x - 2)^2 - 1$
C. $y = (x + 2)^2 + 1$
D. $y = (x - 2)^2 + 1$



Signal. Read the vertex $(-2, -1)$. $(x - 2)^2$ = sign of h reversed, the slip the pre-session also warns about.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach perfect squares and halving with me. **3–4** → straight to completing the square; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.